Beyond the Boundaries -
Lombardy - Italy

Our proposal, which is located in Romano di Lombardia, is guided by the following research question:
"What are the values of the landscape that have been lost in the past, that once replaced could rebalance the interactions of the infrastructure in the territory?"

This question shaped the way we looked at the site, not only through its physical structures but also through the hidden values embedded in water systems, agricultural patterns, and everyday perceptions. **Our approach was to go beyond boundaries and weave together a new network of land and water.**

The first step was to retrace the history of the territory. **The construction of the BreBeMi motorway and the railway fragmented large portions of the land,** interrupting the historical continuity of fields, canals, and settlements. At the same time, features such as the **fontanili, productive fields, and irrigation canals remained as key structuring elements** that once defined the identity of the plain. Recognizing these elements helped us understand what had been lost and what could be brought back.

The existing infrastructures were detailed mapped. The major rivers Adda, Serio, and Oglio, the historic Fosso Bergamasco canal, and the national bike lane AIDA positioned our site within a larger regional system. Within this framework, **the geological line of the fontanili was identified as a critical feature**, since these springs, shaped over centuries by human interventions, provided fresh water for irrigation and created semi-natural flooded landscapes in winter. Just as important was *the role of perception:* from the railway, the landscape reveals itself as a sequence of fragmented boundaries and residual spaces where canals, fields, settlements, and infrastructures meet without clarity. This perception guided our focus towards infraspaces as a field of intervention.

We studied these infraspaces and diffuse borders carefully. **Some were squeezed between canals and productive fields, others lay beside settlements or underneath raised roads, while others stretched along the railway.** Using graphic codes and annotations, we defined their boundaries and recognized them as more than empty margins: they could become active components of a new landscape strategy. With this foundation, we turned to **water systems as the central theme.** Drainage from the BreBeMi and railway currently flows without management, often **polluted and disconnected from the natural hydrology.**

Our proposal introduces **widened drainage canals, bioswales, and detention ponds to collect, filter, and reintegrate these flows.** In one selected infraspace, we developed a detailed design and a three-dimensional model. Here, drainage water from BreBeMi and the railway is gathered through inlet and outlet canals, purified through vegetation, and stored in detention ponds. Alongside this, a separate pond is dedicated to naturalizing water coming from irrigation canals linked to the fontanili, reinforcing their ecological and cultural function.

The design also integrates reforestation strategies with layered vegetation, creating a mosaic of herbaceous plants, shrubs, and trees that improve biodiversity and soil stability. The 3D visualization shows how these interventions transform what was once a neglected space into a functioning hydrological and ecological system.

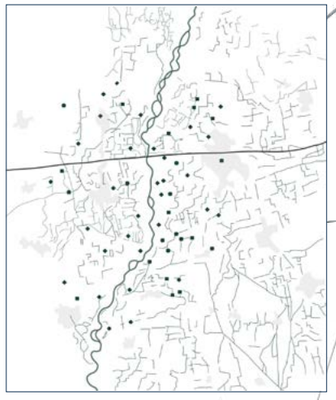
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BEYOND THE BOUNDARIES

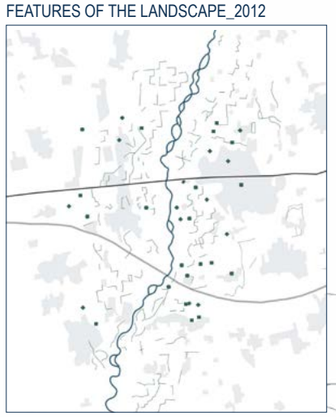
Weaving a new network through land and water

TERRITORY: AN HISTORICAL REVIEW

FEATURES OF THE LANDSCAPE_1975



FEATURES OF THE LANDSCAPE_2012



Elements of the land, like strips of forest located peripherally have dissapeared over time, as a result of the agriculture industralization process.

LEGEND
- Fontanili
- Rooms
- Productive fields
- Settlements
- Primary viability
- Canals

0 1 2 km

RESEARCH QUESTION:

"What are the values of the landscape that have been lost in the past, that once replaced could rebalance the interactions of the infrastructure in the territory?"

LAYERS OF PREEXISTENT VALUES

GEOLOGICAL BORDER OF FONTANILI PRESENCE



The geological line of the springs:
Around of the Serio river, this represents a geomorphological phenomenon linked to the presence of impermeable shallow stratification.

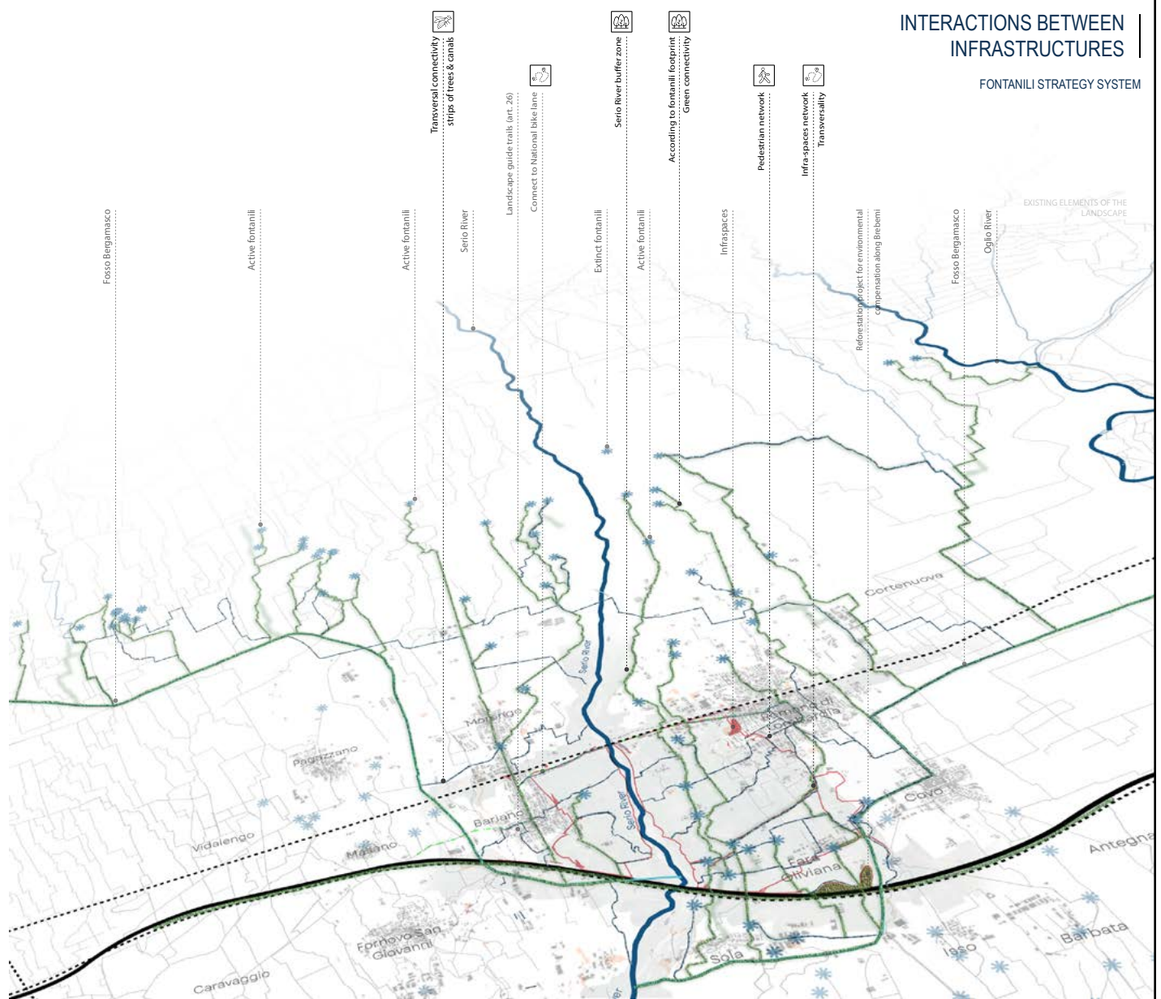
FONTANILI IN THE LAND



The springs represent a peculiar element in the land for the fresh water and for the particular "semi-natural" origin.

INTERACTIONS BETWEEN INFRASTRUCTURES

FONTANILI STRATEGY SYSTEM



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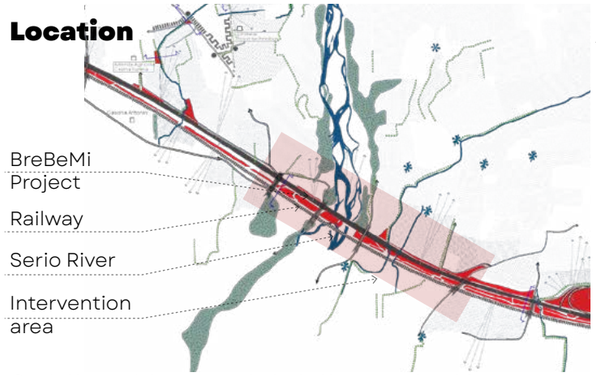
Mining Area
Botanic Garden
Trail Fontana del Carme
Fontana El carmen
Station Park
Il Gardino di Bariano
Serio River
International School
Mondi Italia
Train Station
Education Institutions
Romano Hospital
Historic Centre Romano di Lombardia
Scuola Prima
Piscina Communale
Club di equitazione
Campo sportivo
Farm Factory
Campo Sportivo Parrocchiale
LDP Italia Thrust technology
Azienda Agricola Casina Aurelia
Cascina Antonini
Biogas
Fontanili
Sportivo Comunale
Fara Olivana
Logistic
Cascina Trobbiate
B1
B1'
B2
B2'
B3
B3'
C1
C1'
C2
C2'
C3
C3'
A1
A1'
A2
A2'
A3
A4
A4'
D1
D1'
D2
D2'

A mapping process was carried out to identify all the elements that contributed to the border definition in the area of intervention.

- Residencial
- Institutional
- Industrial
- Tunnels
- Points of decision
- Landmarks
- Principal roads
- Secondary roads Institutional
- Buildings border
- Active Fontanili
- Inactive Fontanili
- Extinct Fontanili
- Local roads
- Railway
- Tree lined strips
- Canals
- Sections

0 100 200 m

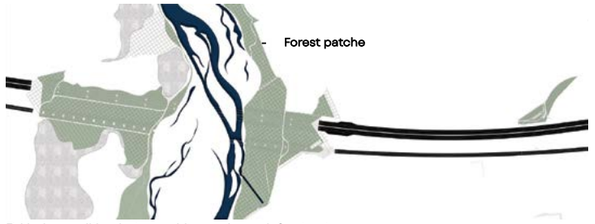
Location



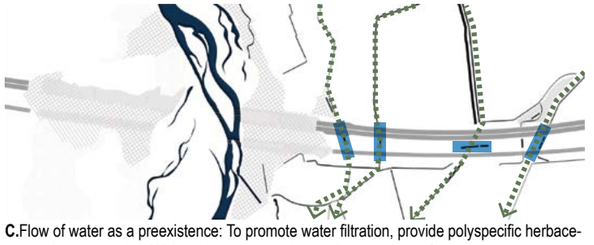
- Sustainable water drainage systems that involve blue, green and gray infrastructure.
- Biodiversity and water resilience.



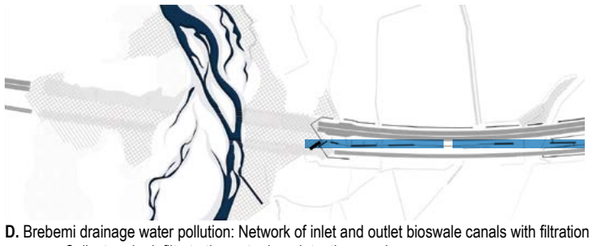
A.Identification of existing condition of infra-spaces and interaction with blue,green and gray infrastructures.



B.Node condition between blue and gray infrastructures.
Actions: -Reforestation plan including shrubs, bushes and trees species. -Connecting existing forest patches.



C.Flow of water as a preexistence: To promote water filtration, provide polyspecific herbaceous and trees species. Widen the water crossings in between the infrastructure.



D. Brebemi drainage water pollution: Network of inlet and outlet bioswale canals with filtration purpose. Collect and reinfiltrate the water in a detention pond.

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A'

B'

B

LEGEND

1 Brebemi Motorway
2 Railway
3 Gray water canals
4 Opening of gray water canals
5 Irrigation canals
6 Detention pond
7 Existing Brebemi compensation plan.
8 Proposed reforestation
9 Proposed path

0 40 m

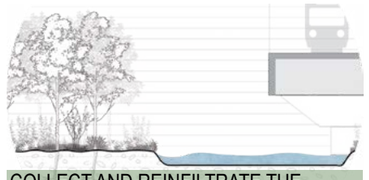
DETAILS OF THE NETWORKS

Between BreBeMi motorway and railway featuring layered vegetation creating a biodiverse space where linear passage creates access for connection to the Serio River.



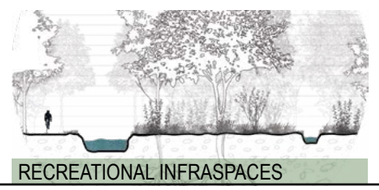
FORESTATION CROSSINGS

Gray water from both infrastructures is filtered through canals and a detention pond, which connects to the wider water system.



COLLECT AND REINFILTRATE THE WATER

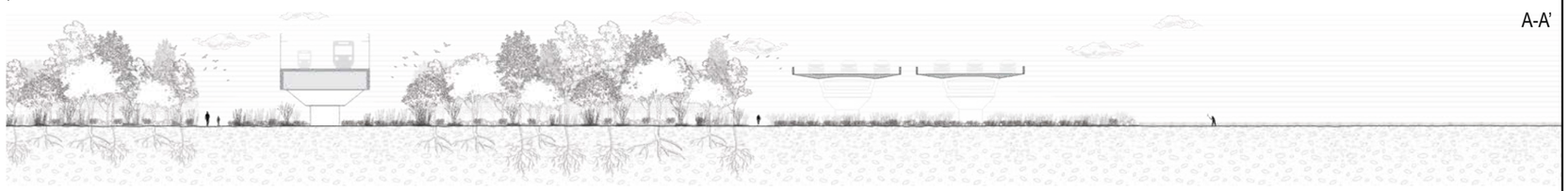
Openings in the embankments filter gray water through the landscape, while reforestation blends ecological restoration with recreational spaces in the corridor.



RECREATIONAL INFRASPACES

TRANSECTS OF THE NETWORK (scale 1:500)

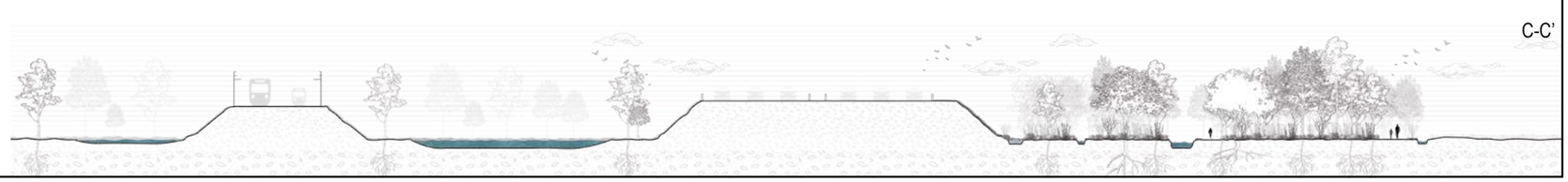
A-A'



B-B'

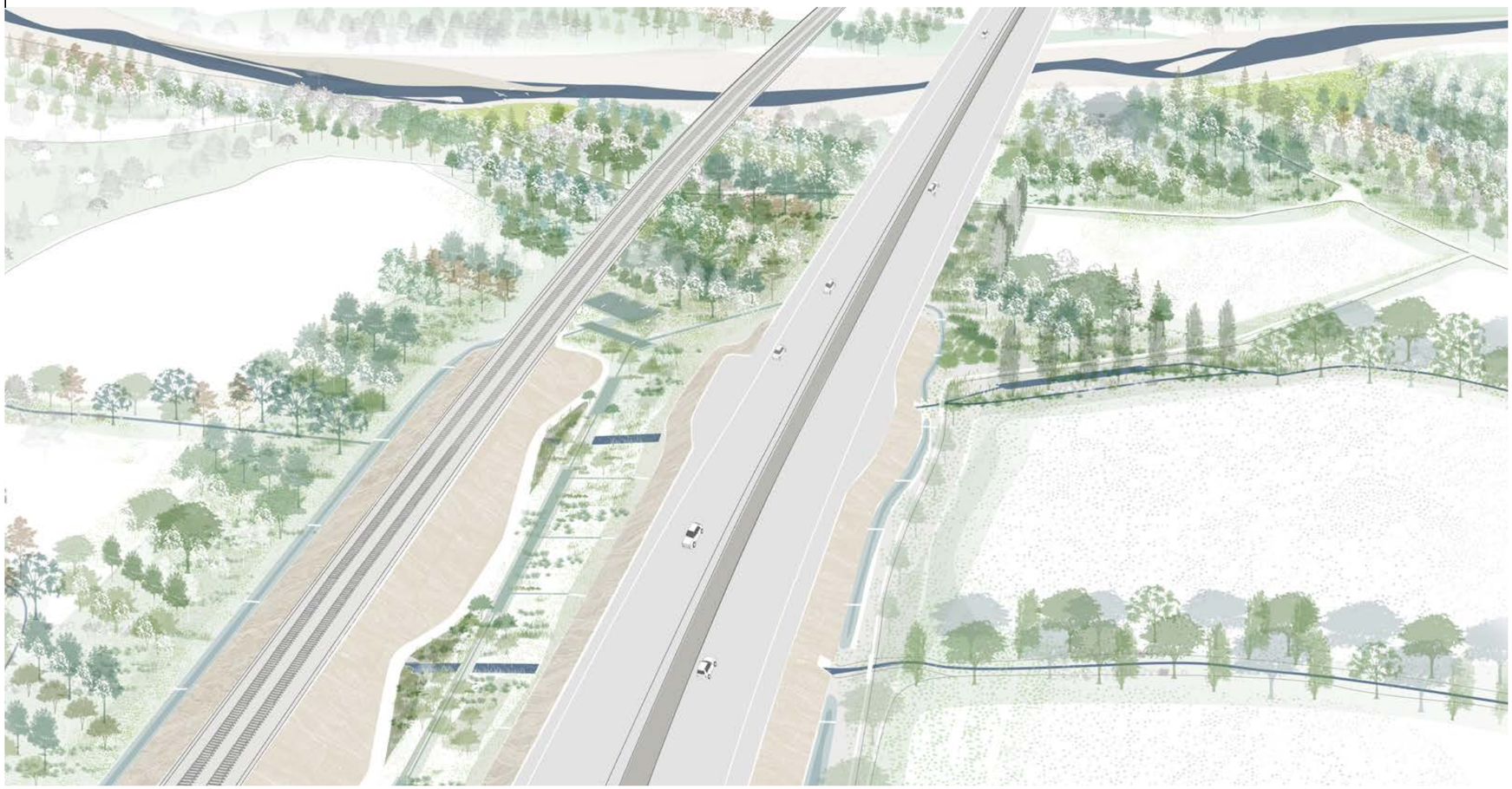


C-C'

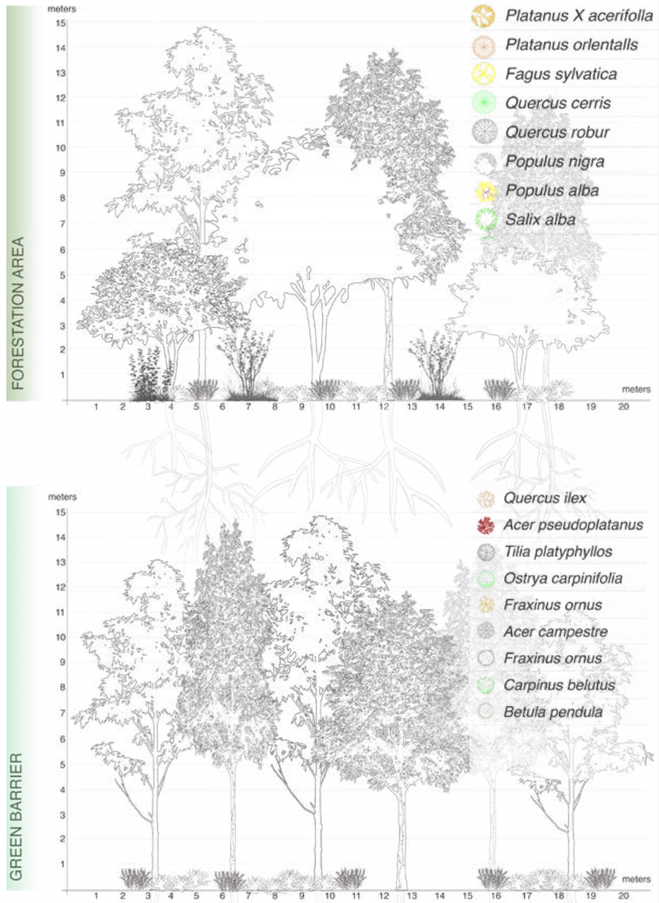


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BIRD-EYE PERSPECTIVE OF THE LAND



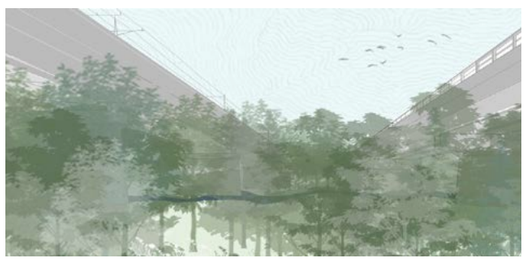
VEGETATION TYPOLOGY



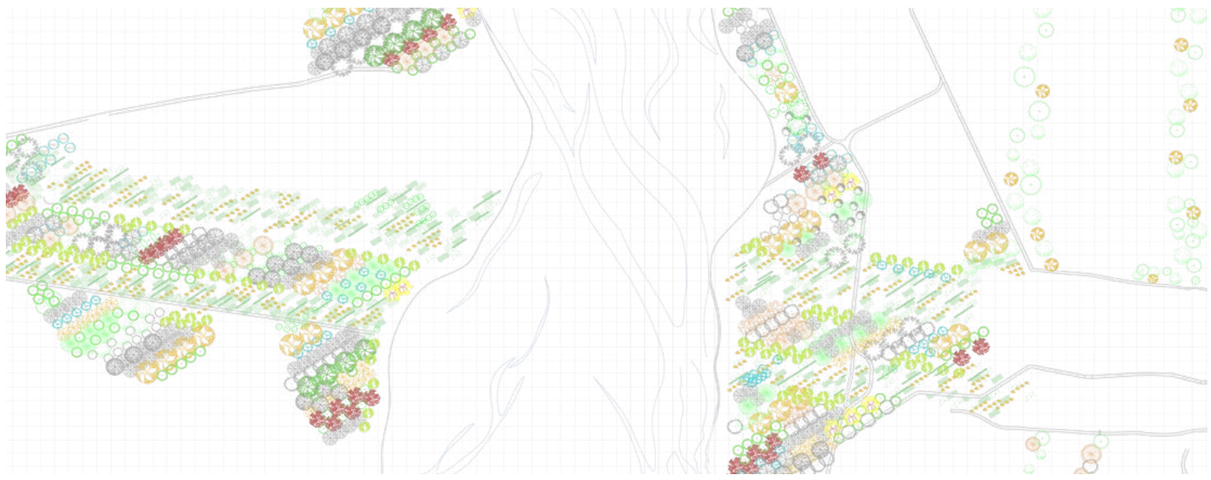
PERCEPTION FROM BREBEMI



PERCEPTION FROM UNDER INFRASTUCTURE



PLANTING PLAN



Planting typologies use native species by function—layered forests, green barriers, bioswales, and detention ponds—to enhance ecological resilience, manage water, and strengthen landscape connectivity.

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